

Computational Microelectronics

Lecture 9 (Pre-recorded)

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L9

Issues

- We have written a code to calculate I_D .
 - ~~1) Electron continuity equation is analytically solved.~~
 - 2) MOS equation and continuity equation are separately treated.
 - 3) Mobility is assumed to be constant.
 - 4) Short-channel MOSFET
 - 5) Source/drain regions
 - 6) Transient effect
 - 7) Multiple devices (Can we calculate the VTC of a CMOS inverter?)

Back to the electron continuity

- The real problem of interest

- The electron continuity $\frac{d}{dx} \left(Q_n(\phi_s, \phi_n) \mu_n \frac{d\phi_n}{dx} \right) = 0$.

- At $x = x_i$,

$$Q_n(\phi_s(x_i), \phi_n(x_i)) \mu_n \frac{\phi_n(x_{i+1}) - \phi_n(x_i)}{\Delta x} - Q_n(\phi_s(x_{i-1}), \phi_n(x_{i-1})) \mu_n \frac{\phi_n(x_i) - \phi_n(x_{i-1})}{\Delta x} = 0$$

- Dividing by $\frac{\mu_n}{\Delta x}$,

$$Q_n(\phi_s(x_i), \phi_n(x_i)) \phi_n(x_{i+1}) - Q_n(\phi_s(x_i), \phi_n(x_i)) \phi_n(x_i) - Q_n(\phi_s(x_{i-1}), \phi_n(x_{i-1})) \phi_n(x_i) + Q_n(\phi_s(x_{i-1}), \phi_n(x_{i-1})) \phi_n(x_{i-1}) = 0$$

Simplify the notation.

- $F_{\phi_{n,i}} = Q_{n,i}\phi_{n,i+1} - (Q_{n,i} + Q_{n,i-1})\phi_{n,i} + Q_{n,i-1}\phi_{n,i-1} = 0$

– Entries of the Jacobian matrix

$$\frac{\partial F_{\phi_{n,i}}}{\partial \phi_{n,i+1}} = Q_{n,i}$$

$$\frac{\partial F_{\phi_{n,i}}}{\partial \phi_{n,i}} = \frac{\partial Q_{n,i}}{\partial \phi_{n,i}} (\phi_{n,i+1} - \phi_{n,i}) - (Q_{n,i} + Q_{n,i-1})$$

$$\frac{\partial F_{\phi_{n,i}}}{\partial \phi_{n,i-1}} = - \frac{\partial Q_{n,i-1}}{\partial \phi_{n,i-1}} (\phi_{n,i} - \phi_{n,i-1}) + Q_{n,i-1}$$

$$\frac{\partial F_{\phi_{n,i}}}{\partial \phi_{s,i}} = \frac{\partial Q_{n,i}}{\partial \phi_{s,i}} (\phi_{n,i+1} - \phi_{n,i})$$

$$\frac{\partial F_{\phi_{n,i}}}{\partial \phi_{s,i-1}} = - \frac{\partial Q_{n,i-1}}{\partial \phi_{s,i-1}} (\phi_{n,i} - \phi_{n,i-1})$$

Expression for $\frac{\partial Q_n}{\partial \phi_s}$

- Recall that

$$Q_n(x) = -q \int_{\delta}^{\phi_s} \frac{\frac{n_{int}^2}{N_a} \exp\left(\frac{q(\bar{\phi} - \phi_n)}{k_B T}\right)}{E_z(\bar{\phi}, \phi_n)} d\bar{\phi}$$

- It is clear that

$$\frac{\partial Q_n}{\partial \phi_s} = -q \frac{\frac{n_{int}^2}{N_a} \exp\left(\frac{q(\phi_s - \phi_n)}{k_B T}\right)}{E_z(\phi_s, \phi_n)}$$

- Then, how about $\frac{\partial Q_n}{\partial \phi_n}$? For that purpose, we introduce a small $\delta\phi_n$.

Expression for $\frac{\partial Q_n}{\partial \phi_n}$

- With a small $\delta \phi_n$,

$$\begin{aligned} \delta Q_n &= -q \int_{\delta}^{\phi_s} \frac{n_{int}^2}{N_a} \exp\left(\frac{q(\bar{\phi} - \phi_n)}{k_B T}\right) \left(-\frac{q \delta \phi_n}{k_B T}\right) d\bar{\phi} \\ &\quad - q \int_{\delta}^{\phi_s} \frac{n_{int}^2}{N_a} \exp\left(\frac{q(\bar{\phi} - \phi_n)}{k_B T}\right) \left(-\frac{1}{E_z} \frac{\partial E_z}{\partial \phi_n} \delta \phi_n\right) d\bar{\phi} \end{aligned}$$

- The first term is easily related to Q_n .
- The second term needs additional numerical integration.

Constructing the Jacobian matrix

- Cost

- $\frac{\partial Q_n}{\partial \phi_s}$ can be readily calculated.

- $\frac{\partial Q_n}{\partial \phi_n}$ has two terms. One term is easy to calculate.

- The other one requires additional numerical integration. It can be done just like Q_n integration.

- Overall,

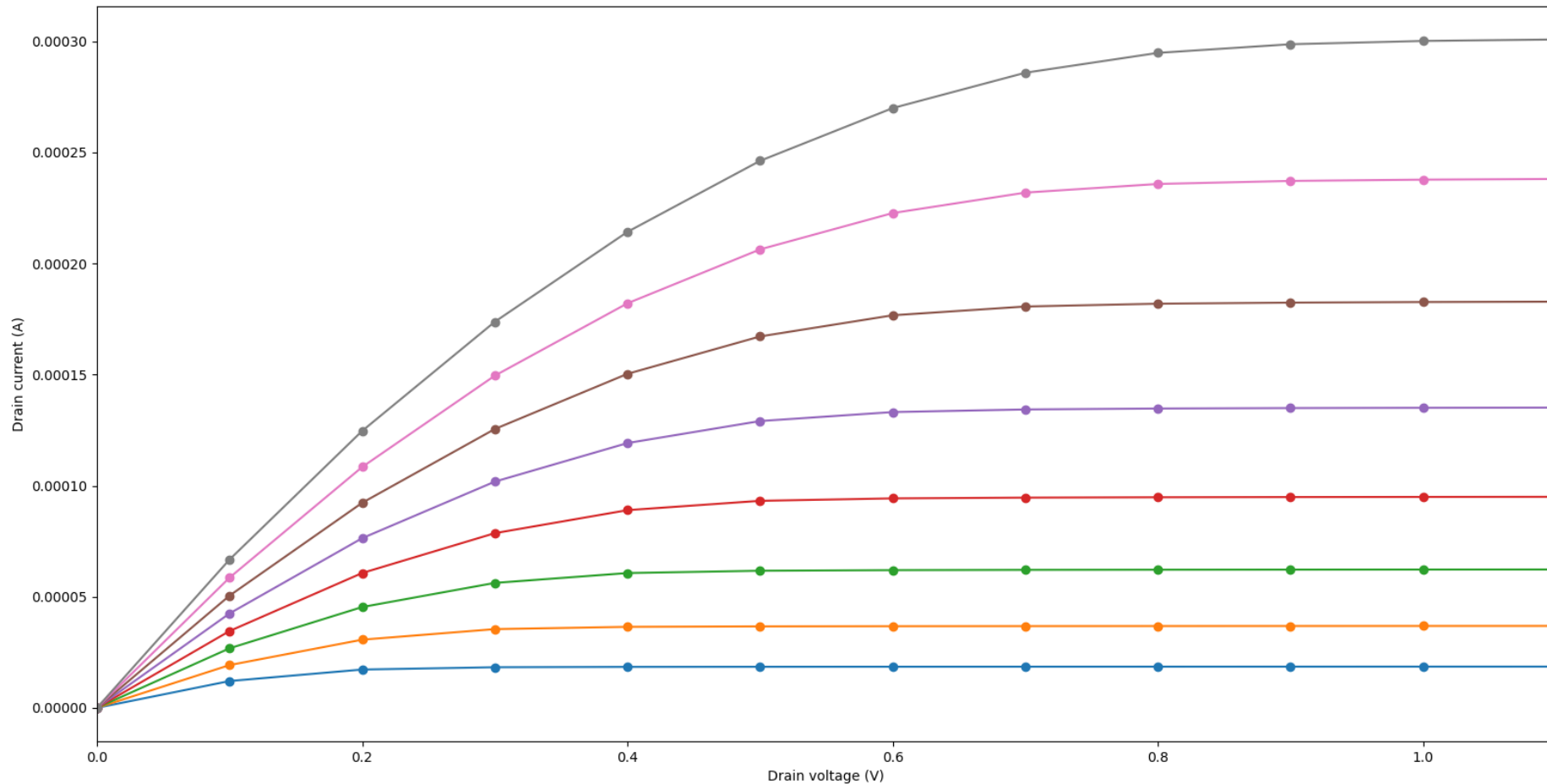
- If we are careful, we can implement the accurate Jacobian matrix.

Which entry is important?

- Test result @ $V_G = 1.1 \text{ V}$
 - Completely decoupled $\rightarrow$ Converged only up to $V_D = 0.1 \text{ V}$
 - Add $\frac{\partial}{\partial \phi_n} F_{\phi_s}$ to the MOS equation $\rightarrow$ Converged up to $V_D = 0.4 \text{ V}$
 - Add $\frac{\partial Q_n}{\partial \phi_s}$ to the electron continuity equation $\rightarrow$ Converged up to $V_D = 0.4 \text{ V}$ (No improvement)
 - Add $\frac{\partial Q_n}{\partial \phi_n}$ to the electron continuity equation $\rightarrow$ Converged up to $V_D = 1.1 \text{ V}$
- Accurate Jacobian matrix is very important.

Calculate the I_D - V_D curve, again.

- Already done in Homework#7
 - Improved convergence!



Thank you!